Design

QUANTUM DOTS: An Emerging Field Of Technology

DEEPSHIKHA SHUKLA

A central theme in nanotechnology, quantum dots can tune their band gap, which enables these to control the frequency of light emission and absorbance

Quantum dots (QDs) are semiconducting particles that are a few nanometres in size. Due to their small size, these can confine electrons' motions. These have the ability to tune their band gap, which enables these to control the frequency of light emission and absorbance. After absorbing a photon of light, these emit a photon of longer wavelength. Wavelength of the emitted photon can be controlled by controlling the size of QDs. Hence, it is possible to control their properties (electrical and optical) according to their use.

A full range of QDs can be manufactured, each with a narrow, distinct emission spectrum by manipulating the colour of light emitted from these. QDs emit light of a particular colour after being illuminated by light. The colour depends on the size of the nanoparticle, that is, difference between conduction band (CB) and valence band (VB). The smaller the nanoparticle, the larger the difference between CB and VB, resulting in a deeper blue colour. For a larger nanoparticle, the difference between CB and VB is lower, which shifts the glow towards red.

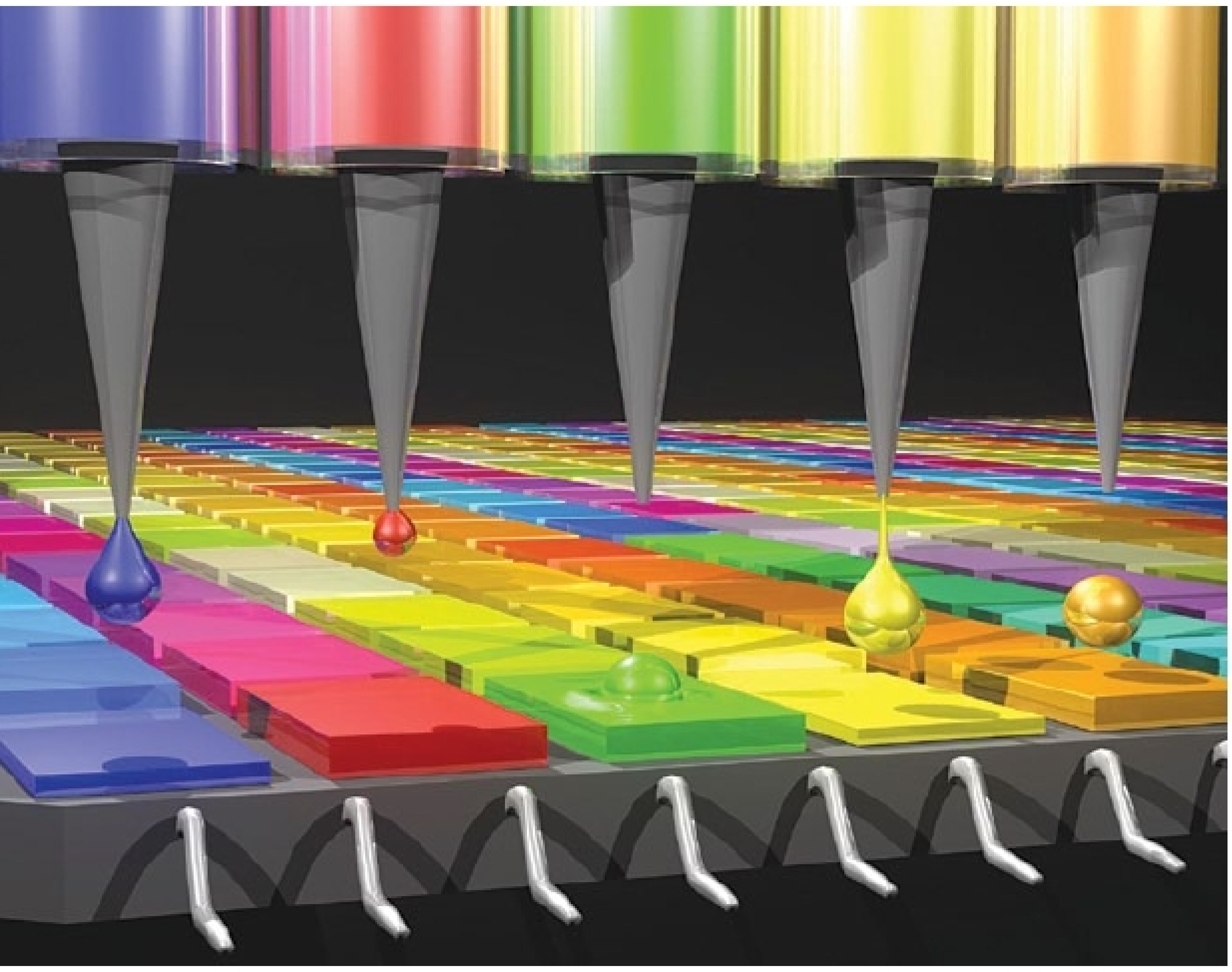

Fig. 1: Quantum dot filters absorb different wavelengths of light depending on their size and composition (Credit: O'Reilly Science Art)

Recently, a NASA technologist has teamed with the inventor of a new nanotechnology that could transform the way space scientists build spectrometers. It is an important device used by all scientific disciplines to measure the properties of light emanating from astronomical objects, including Earth.

Absorption spectrometers measure absorption of light as a function of frequency or wavelength due to its interaction with a sample, such as atmospheric gases. This technology allows instrument developers to generate nearly an unlimited number of different dots, and have precise control over what each dot absorbs. It helps them customise instruments to observe many different bands with high-spectral resolution. This emerging technology could give scientists a more flexible, cost-effective approach for developing spectrometers.

Working of QD technology

When an electron absorbs a photon, it rises to CB from VB. In doing so it leaves a hole in VB, the electron and hole pair together called exciton. Average distance between the electron and hole is known as Bohr Radius. When the size of the semiconductor reduces in comparison to Bohr Radius, it is called QD. After a certain time period, the electron falls back to VB and combines with the hole. This recombination process releases a photon of frequency that depends on band gap energy.

As the size of the crystal decreases, band gap increases. The greater the difference in the energy between highest VB and lowest CB, the more the energy required to excite the dot and, simultaneously, the more the energy released when the crystal returns to VB. This results in higher frequencies of light emitted after excitation of the dot. QDs can be engineered to fluorescence in different wavelengths based on their physical dimensions.